Hadoop Installation Guide

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Note For Windows User

If you're **using Windows**, skip the macOS-specific step below. Continue from **Step 2** of the main guide.

Also, make sure you're using **WSL (Windows Subsystem for Linux)** with **Ubuntu** to run the commands below.

1. Hadoop Installation For MacOS

If you are using macOS, the installation process is slightly different from Linux. Please follow the steps outlined in the guide below:

👉 Hadoop Installation Guide For MacOS

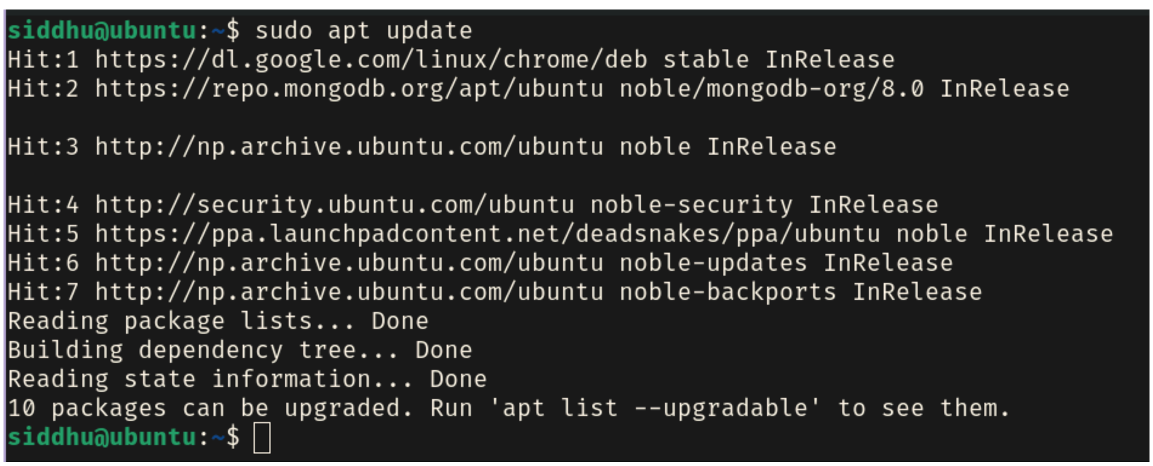
This guide includes:

- Installing Java on macOS
- Configuring environment variables
- Setting up Hadoop and verifying the installation

2. Java Installation

The Hadoop framework is written in Java, and its services require a compatible Java Runtime Environment (JRE) and Java Development Kit (JDK). Use the following command to update your system before initiating a new installation. First thing you want to do is to install updates.

```
~$ sudo apt update
```



Here we are going to install the open jdk 8 package as it contains both runtime and the development key.
Open the terminal and type the following code :

```
~$ sudo apt install openjdk-8-jdk -y
```

```
siddhu@ubuntu:~$ sudo apt install openjdk-8-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libice-dev libsm-dev libxt-dev openjdk-8-jdk-headless
Suggested packages:
  libice-doc libsm-doc libxt-doc openjdk-8-demo openjdk-8-source visualvm
The following NEW packages will be installed:
  libice-dev libsm-dev libxt-dev openjdk-8-jdk openjdk-8-jdk-headless
0 upgraded, 5 newly installed, 0 to remove and 1 not upgraded.
```

Check if the java is installed in the pc.

~$ java -version; javac -version

```
siddhu@ubuntu:~$ java -version; javac -version
openjdk version "1.8.0_442"
OpenJDK Runtime Environment (build 1.8.0_442-8u442-b06~us1-0ubuntu1~24.04-b06)
OpenJDK 64-Bit Server VM (build 25.442-b06, mixed mode)
javac 1.8.0_442
siddhu@ubuntu:~$
```

3. Setting up a dedicated user for Hadoop user

A distinct user improves security and helps us arrange clusters more efficiently.

hdoop : *It is the name of the user account that **you will have to** dedicate to Hadoop.*

3.1 First we want to do is install open SSH on Ubuntu

Install openssh server and client using the following command :

~$ sudo apt install openssh-server openssh-client -y

```
siddhu@ubuntu:~$ sudo apt install openssh-server openssh-client -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-client openssh-server openssh-sftp-server
  ssh-import-id
0 upgraded, 5 newly installed, 0 to remove and 1 not upgraded.
Need to get 1,737 kB of archives.
After this operation, 10.4 MB of additional disk space will be used.
Get:1 http://np.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-cli
ent amd64 1:9.6p1-3ubuntu13.9 [905 kB]
Get:2 http://np.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sft
p-server amd64 1:9.6p1-3ubuntu13.9 [37.3 kB]
Get:3 http://np.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-ser
```

3.2 Creating Hadoop user

1. We are going to use the adduser command to add the user

~$ sudo adduser hdoop

Note : *Username is very crucial part here*

```
siddhu@ubuntu:~$ sudo adduser hdoop
info: Adding user `hdoop' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `hdoop' (1001) ...
info: Adding new user `hdoop' (1001) with group `hdoop (1001)' ...
warn: The home directory `/home/hdoop' already exists.  Not touching this directory.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for hdoop
Enter the new value, or press ENTER for the default
        Full Name []: Siddhartha Shakya
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `hdoop' to supplemental / extra groups `users' ...
info: Adding user `hdoop' to group `users' ...
siddhu@ubuntu:~$
```

2. Let's switch the user to newly created user with the following command

```
~$ su - hdoop
```

```
siddhu@ubuntu:~$ su - hdoop
Password:
hdoop@ubuntu:~$
```

3.3 Enabling Password less SSH for Hadoop user

1. Generating an ssh key pair and define the location to be stored

```
~$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
```

Note : *"-f ~/.ssh/id_rsa" specifies the file location where the private key will be saved (in this case, ~/.ssh/id_rsa).*

```
hdoop@ubuntu:~$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
/home/hdoop/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Your identification has been saved in /home/hdoop/.ssh/id_rsa
Your public key has been saved in /home/hdoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:clLMC1irLWYN6cnEiCUrlJED9Md5ilA0a5c9nh5Kd0Y hdoop@ubuntu
The key's randomart image is:
+---[RSA 3072]----+
|=+*+   .         |
|.Ooo++++         |
|+.o+**o++E       |
|. o+*+o+.        |
|   .Oo==So       |
|   o..+++        |
|     . .         |
|                 |
|                 |
+----[SHA256]-----+
hdoop@ubuntu:~$
```

2. Using the cat command to store the public key as authorized_key in the ssh directory by the following command

```
~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

3. Setting the permissions for the user with the chmod command

```
~$ chmod 0600 ~/.ssh/authorized_keys
```

```
hdoop@ubuntu:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
hdoop@ubuntu:~$ chmod 0600 ~/.ssh/authorized_keys
hdoop@ubuntu:~$
```

This is the correct permission setting for the authorized_keys file to ensure it can be read and written by the owner (you) but not by anyone else. This permission is required for SSH to work correctly, as it prevents unauthorized users from modifying the authorized_keys file.

4. Now ssh the local host with the following command

```
~$ ssh localhost
```

```
hdoop@ubuntu:~$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:/k9j/cB1NQVeFAQdE1x0NOUXku8ZmXQCnyHs0ozrMbo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-19-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

16 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

hdoop@ubuntu:~$
```

4. Downloading and installing Hadoop locally

4.1 Downloading the Hadoop file

1. We have to specify the version we want to download properly. **Hadoop 3.2.4** is recommended.

```
~$ wget https://downloads.apache.org/hadoop/common/hadoop-3.2.4/hadoop-3.2.4.tar.gz
```

```
hdoop@ubuntu:~$ wget https://downloads.apache.org/hadoop/common/hadoop-3.2.4/hadoop-3.2.4.tar.gz
--2025-03-31 08:40:21--  https://downloads.apache.org/hadoop/common/hadoop-3.2.4/hadoop-3.2.4.tar.gz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 492368219 (470M) [application/x-gzip]
Saving to: ‘hadoop-3.2.4.tar.gz’

hadoop-3.2.4.tar.gz                          0%[
```

2. Now with the tar we will have to extract the Hadoop for initializing the installation

```
~$ tar xzf hadoop-3.2.4.tar.gz
```

```
hdoop@ubuntu:~$ tar xzf hadoop-3.2.4.tar.gz
hdoop@ubuntu:~$ ls
hadoop-3.2.4  hadoop-3.2.4.tar.gz  snap
hdoop@ubuntu:~$
```

All the Hadoop files are now located in **hadoop-3.2.4**

4.2 Adding User to the sudoers group for privileges

1. To edit the configuration file we must provide proper root privileges. So we have to add the newly created user to the admin group. **Switch to the main root user**

```
~$ su - siddhu
```

Note : *you must replace “siddhu” with your* ***own main root username****.*

2. Run the following command

```
~$ sudo usermod -aG sudo hdoop
```

```
siddhu@ubuntu:~$ sudo usermod -aG sudo hdoop
[sudo] password for siddhu:
siddhu@ubuntu:~$ groups hdoop
hdoop : hdoop sudo users
siddhu@ubuntu:~$
```

4.3 Configuring Hadoop Environment Variables

Now switch back to your Hadoop user with the following code

```
~$ su - hdoop
```

1. To check if the Hadoop user is in the sudoers. (Checking for root privileges).

```
~$ sudo whoami
```

```
hdoop@ubuntu:~$ sudo whoami
[sudo] password for hdoop:
root
hdoop@ubuntu:~$
```

2. We have to first edit the .bashrc file.

```
~$ sudo nano .bashrc
```

3. After the bashrc file opens, add the following code. Make sure you check your variable accordingly. **Navigate to the end of the file and add the following code.**

```
#Hadoop Related Options

export HADOOP_HOME=/home/hdoop/hadoop-3.2.4

export HADOOP_INSTALL=$HADOOP_HOME

export HADOOP_MAPRED_HOME=$HADOOP_HOME

export HADOOP_COMMON_HOME=$HADOOP_HOME

export HADOOP_HDFS_HOME=$HADOOP_HOME

export YARN_HOME=$HADOOP_HOME

export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native

export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin

export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
```

```
# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

#Hadoop Related Options

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/home/hdoop/hadoop-3.2.4
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/sbin:$HADOOP_HOME/bin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/nativ"
```

4. Now we must apply the changes to the currently running environment. To do that type the following command.

```
~$ source ~/.bashrc
```

```
hdoop@ubuntu:~$ sudo nano .bashrc
hdoop@ubuntu:~$ source ~/.bashrc
hdoop@ubuntu:~$
```

5. Now we have to edit the Hadoop-env.sh file.

- Here first we have to know the location of the file where our java is installed.

```
~$ which javac
```

```
hdoop@ubuntu:~$ which javac
/usr/bin/javac
hdoop@ubuntu:~$
```

- Let's find the open JDK directory

```
~$ readlink -f /usr/bin/javac
```

```
hdoop@ubuntu:~$ readlink -f /usr/bin/javac
/usr/lib/jvm/java-8-openjdk-amd64/bin/javac
hdoop@ubuntu:~$
```

Copy the path upto amd64. My copied path :

```
/usr/lib/jvm/java-8-openjdk-amd64
```

6. Now lets open the **hadoop –env.sh** file

```
~$ sudo nano $HADOOP_HOME/etc/hadoop/hadoop-env.sh
```

7. Search for **JAVA_HOME** and uncomment it (delete the #) and paste the copied location.

```
###
# Generic settings for HADOOP
###

# Technically, the only required environment variable is JAVA_HOME.
# All others are optional.  However, the defaults are probably not
# preferred.  Many sites configure these options outside of Hadoop,
# such as in /etc/profile.d

# The java implementation to use. By default, this environment
# variable is REQUIRED on ALL platforms except OS X!
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

# Location of Hadoop.  By default, Hadoop will attempt to determine
# this location based upon its execution path.
# export HADOOP_HOME=

# Location of Hadoop's configuration information.  i.e., where this
# file is living. If this is not defined, Hadoop will attempt to
# locate it based upon its execution path.
#
```

8. Now let's edit the **core-site.xml** file.

```
~$ sudo nano $HADOOP_HOME/etc/hadoop/core-site.xml
```

9. At the end of the file add the following configuration.

```
<configuration>
<property>
    <name>hadoop.tmp.dir</name>
    <value>/home/hdoop/tmpdata</value>
</property>
<property>
    <name>fs.default.name</name>
    <value>hdfs://127.0.0.1:9000</value>
</property>
</configuration>
```

```
GNU nano 7.2                    /home/hdoop/hadoop-3.2.4/etc/hadoop/core-site.xml *
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

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  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
      <name>hadoop.tmp.dir</name>
      <value>/home/hdoop/tmpdata</value>
</property>
<property>
      <name>fs.default.name</name>
      <value>hdfs://127.0.0.1:9000</value>
</property>
</configuration>
```
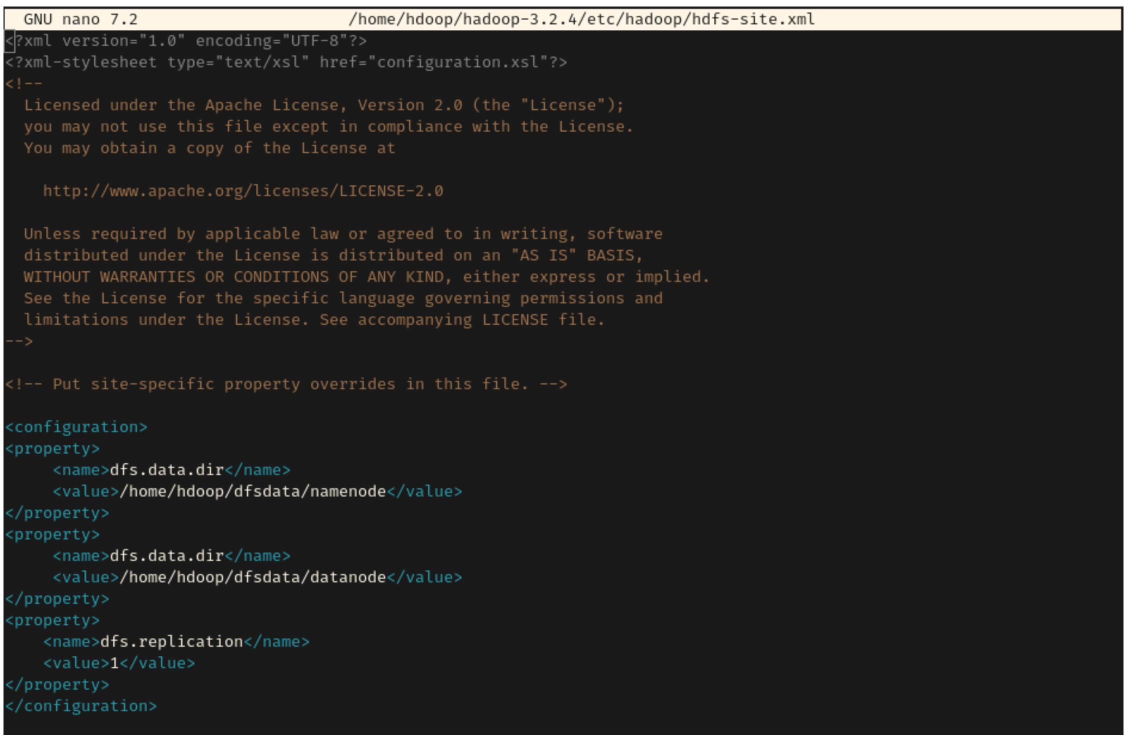

10. Now we have to edit the **hdfs-edit.xml** file. To open the file type the following command.

```
~$ sudo nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
```

11. Add the following command in the file.

```
<configuration>
<property>
    <name>dfs.data.dir</name>
    <value>/home/hdoop/dfsdata/namenode</value>
</property>
<property>
    <name>dfs.data.dir</name>
    <value>/home/hdoop/dfsdata/datanode</value>
</property>
<property>
   <name>dfs.replication</name>
   <value>1</value>
</property>
</configuration>
```



```
  GNU nano 7.2                                  /home/hdoop/hadoop-3.2.4/etc/hadoop/hdfs-site.xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
     <name>dfs.data.dir</name>
     <value>/home/hdoop/dfsdata/namenode</value>
</property>
<property>
     <name>dfs.data.dir</name>
     <value>/home/hdoop/dfsdata/datanode</value>
</property>
<property>
    <name>dfs.replication</name>
    <value>1</value>
</property>
</configuration>
```
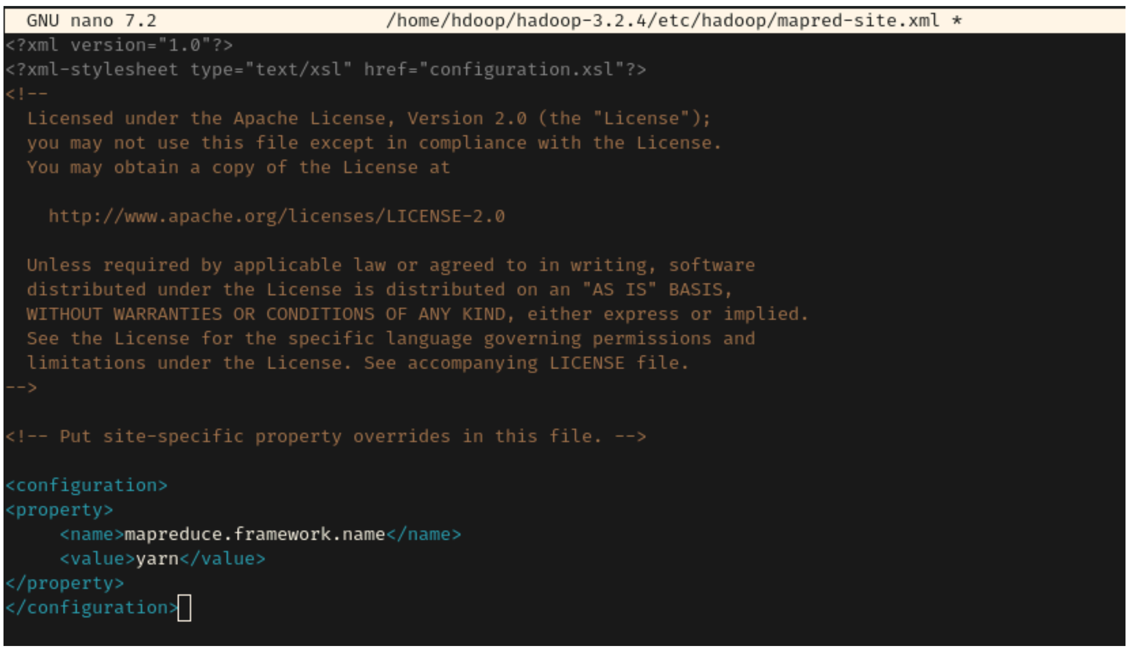

12. Edit the **mapred-site.xml**.

```
~$ sudo nano $HADOOP_HOME/etc/hadoop/mapred-site.xml
```

13. Add the following configuration at last of the file

```
<configuration>
<property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
</property>
</configuration>
```



```
  GNU nano 7.2                    /home/hdoop/hadoop-3.2.4/etc/hadoop/mapred-site.xml *
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
     <name>mapreduce.framework.name</name>
     <value>yarn</value>
</property>
</configuration>
```

14. Editing **yarn-site.xml** file

```
~$ sudo nano $HADOOP_HOME/etc/hadoop/yarn-site.xml
```

15. Add the following configuration in the file.

```
<configuration>
<property>
     <name>yarn.nodemanager.aux-services</name>
     <value>mapreduce_shuffle</value>
</property>
<property>
     <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
     <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
     <name>yarn.resourcemanager.hostname</name>
     <value>127.0.0.1</value>
</property>
<property>
     <name>yarn.acl.enable</name>
     <value>0</value>
</property>
<property>
     <name>yarn.nodemanager.env-whitelist</name>
<value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HA
DOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HO
ME,HADOOP_MAPRED_HOME</value>
</property>
<property>
   <name>yarn.resourcemanager.webapp.address</name>
   <value>0.0.0.0:8088</value>
</property>
</configuration>
```

```
GNU nano 7.2                                   /home/hdoop/hadoop-3.2.4/etc/hadoop/yarn-site.xml *
<?xml version="1.0"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<property>
      <name>yarn.nodemanager.aux-services</name>
      <value>mapreduce_shuffle</value>
</property>
<property>
      <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
      <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
      <name>yarn.resourcemanager.hostname</name>
      <value>127.0.0.1</value>
</property>
<property>
      <name>yarn.acl.enable</name>
      <value>0</value>
</property>
<property>
      <name>yarn.nodemanager.env-whitelist</name>
<value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
</property>
</configuration>
```

16. We now have to format hdfs name node. To do that type the following command.

```
~$ hdfs namenode -format
```

```
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManagerSafeMode: dfs.namenode.safemode.threshold-pct = 0.9990000128
746033
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManagerSafeMode: dfs.namenode.safemode.min.datanodes = 0
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManagerSafeMode: dfs.namenode.safemode.extension = 30000
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManager: defaultReplication         = 1
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManager: maxReplication             = 512
2025-03-31 10:48:27,037 INFO blockmanagement.BlockManager: minReplication             = 1
2025-03-31 10:48:27,038 INFO blockmanagement.BlockManager: maxReplicationStreams      = 2
2025-03-31 10:48:27,038 INFO blockmanagement.BlockManager: redundancyRecheckInterval  = 3000ms
2025-03-31 10:48:27,038 INFO blockmanagement.BlockManager: encryptDataTransfer        = false
2025-03-31 10:48:27,038 INFO blockmanagement.BlockManager: maxNumBlocksToLog          = 1000
2025-03-31 10:48:27,053 INFO namenode.FSDirectory: GLOBAL serial map: bits=29 maxEntries=536870911
2025-03-31 10:48:27,053 INFO namenode.FSDirectory: USER serial map: bits=24 maxEntries=16777215
2025-03-31 10:48:27,053 INFO namenode.FSDirectory: GROUP serial map: bits=24 maxEntries=16777215
2025-03-31 10:48:27,053 INFO namenode.FSDirectory: XATTR serial map: bits=24 maxEntries=16777215
2025-03-31 10:48:27,062 INFO util.GSet: Computing capacity for map INodeMap
2025-03-31 10:48:27,062 INFO util.GSet: VM type       = 64-bit
2025-03-31 10:48:27,062 INFO util.GSet: 1.0% max memory 3.3 GB = 34.1 MB
```

17. To start a Hadoop cluster type the following command.
    First you have to go to the **sbin** folder which is located at your Hadoop sbin folder.
    Hadoop -> sbin

```
~$ cd hadoop-3.2.4/sbin
```

```
hdoop@ubuntu:~$ cd hadoop-3.2.4/sbin
hdoop@ubuntu:~/hadoop-3.2.4/sbin$ ls
distribute-exclude.sh  mr-jobhistory-daemon.sh  start-dfs.sh         stop-balancer.sh     workers.sh
FederationStateStore   refresh-namenodes.sh     start-secure-dns.sh  stop-dfs.cmd         yarn-daemon.sh
hadoop-daemon.sh       start-all.cmd            start-yarn.cmd       stop-dfs.sh          yarn-daemons.sh
hadoop-daemons.sh      start-all.sh             start-yarn.sh        stop-secure-dns.sh
httpfs.sh              start-balancer.sh        stop-all.cmd         stop-yarn.cmd
kms.sh                 start-dfs.cmd            stop-all.sh          stop-yarn.sh
hdoop@ubuntu:~/hadoop-3.2.4/sbin$
```

~$ ./start-dfs.sh

```
hdoop@ubuntu:~/hadoop-3.2.4/sbin$ ./start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [ubuntu]
hdoop@ubuntu:~/hadoop-3.2.4/sbin$
```

18. Starting the yarn resource manager

~$ ./start-yarn.sh

```
hdoop@ubuntu:~/hadoop-3.2.4/sbin$ ./start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hdoop@ubuntu:~/hadoop-3.2.4/sbin$
```

19. Now check if all the java services are running.

~$ jps

```
hdoop@ubuntu:~/hadoop-3.2.4/sbin$ jps
13858 NameNode
14037 DataNode
15560 Jps
14873 ResourceManager
14266 SecondaryNameNode
15197 NodeManager
hdoop@ubuntu:~/hadoop-3.2.4/sbin$
```

20. Make sure that every above java process is running in the background.

*Note : If you don't see **DataNode** running in the background then follow the below steps:*

- Stop the Hadoop services

```
~$ $HADOOP_HOME/sbin/stop-dfs.sh
```

- Delete the DataNode Storage Directory

```
~$ sudo rm -rf /home/hdoop/dfsdata/datanode
```

- Restart Hadoop Services:

```
~$ $HADOOP_HOME/sbin/start-dfs.sh
```

- Check DataNode Status:

```
~$ jps
```

5. Access Hadoop from Browser

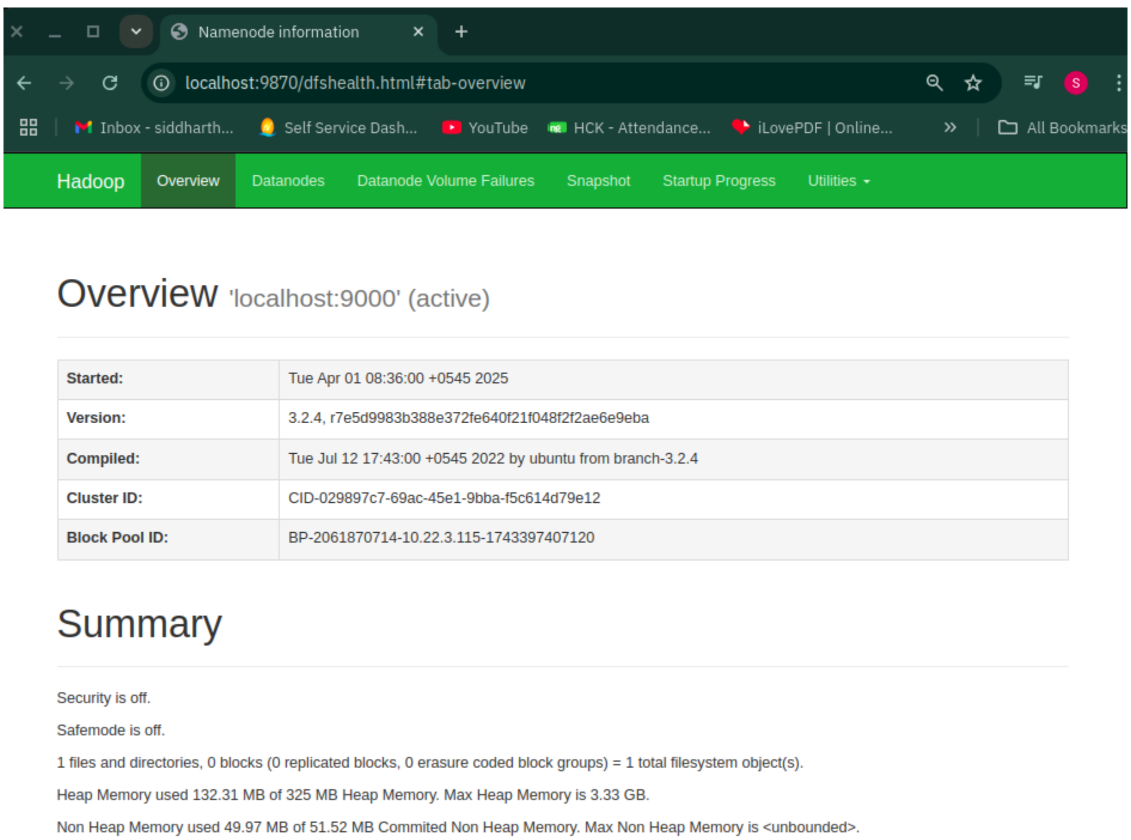
Use your preferred browser and navigate to your localhost URL or IP.

5.1 Access hadoop NameNode UI

The default port number 9870 gives you access to the hadoop NameNode UI:

http://localhost:9870

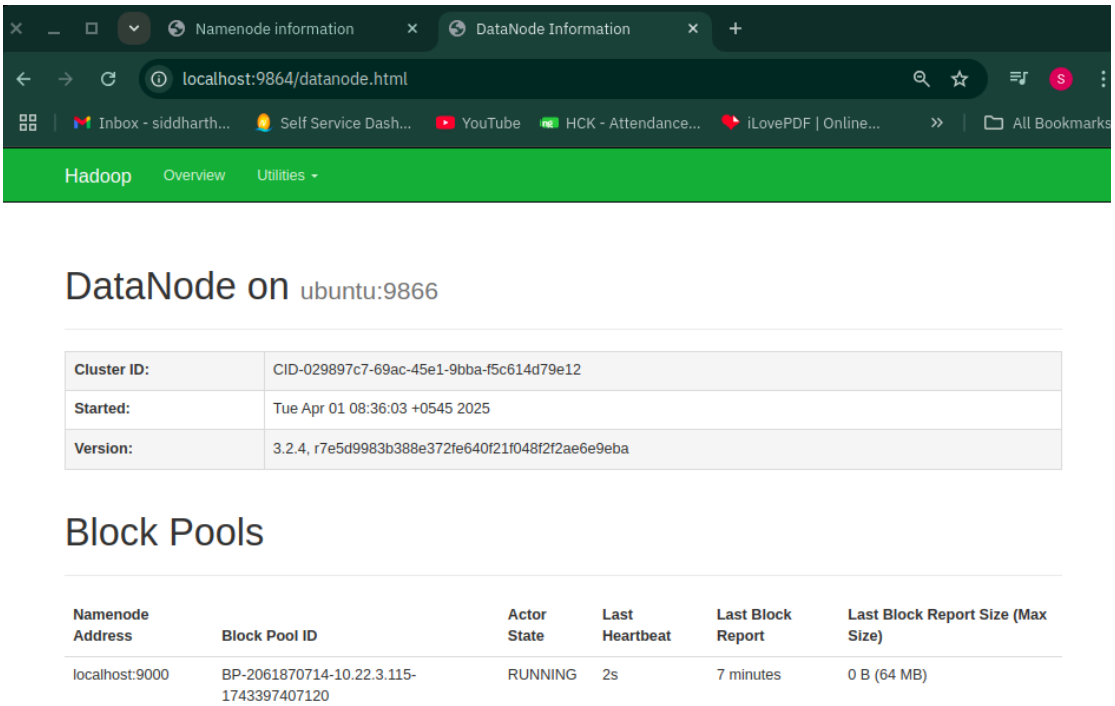
The NameNode user interface provides a comprehensive overview of the entire cluster.



5.2 Access Individual DataNodes

The default port 9864 is used to access individual DataNodes directly from your browser:

```
http://localhost:9864
```

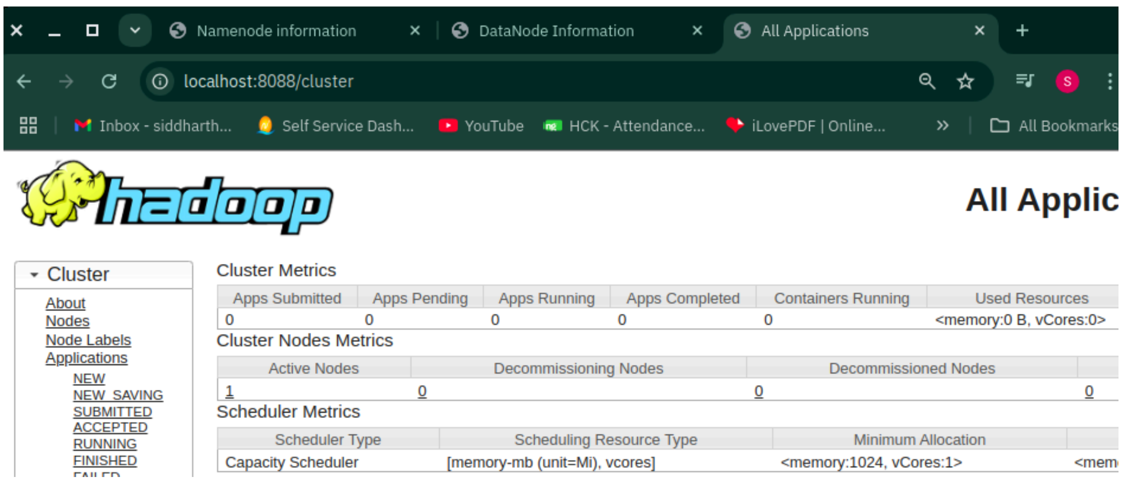


5.2 YARN Resource Manager

The YARN Resource Manager is accessible on port 8088:

```
http://localhost:8088
```

The Resource Manager is an invaluable tool that allows you to monitor all running processes in your Hadoop cluster.



!! Congratulations Hadoop installation Successful **!!**